



WashU VTOL

Washington University in St. Louis

2025–2026 Design-Build-Vertical-Flight Competition

Final Technical Report

March 2, 2026

1 Executive Summary

1.1 Wildfire Response

WashU VTOL developed a fully electric hexacopter VTOL aircraft engineered specifically for the VFS DBVF wildfire response mission. The design prioritizes vertical stability, autonomous payload deployment and pickup, and structural durability while maintaining a competition compliant takeoff weight under 25lb. The platform is engineered to maximize payload throughput, autonomous mission reliability, and vertical landing precision within the competition's 10-minute mission window. Because wildfires expand rapidly during early growth stages, fast and accurate payload delivery is critical. The aircraft supports three operational phases: autonomous transit and precision landing, targeted delivery onto active fire zones, and payload resupply operations. The system was designed to hover over drop zones, minimize reposition time between waypoints, and redeploy with limited ground crew intervention. Autonomous FM-2 and FM-3 capabilities are implemented to pursue autonomy multipliers while preserving manual override for safety compliance.

1.2 Design Overview

A six-rotor configuration was selected to improve hover stability and control authority in variable wind conditions. With an approximate 3:1 thrust-to-weight ratio at maximum payload, the system maintains sufficient margin for vertical takeoff, precision descent, and repeatable drop operations. The hexacopter layout improves positional stability during hover, directly supporting accurate payload deployment and pickup scoring objectives. Propulsion is provided by six T-Motor MN6007 II (320KV) brushless motors each paired with 21×6.3 two-blade propellers. The aircraft is designed such that hover occurs near 50% throttle, staying within efficient motor operation ranges.

The propulsion system is powered by a dedicated 6S 12,000mAh battery, while avionics and payload systems are powered by a separate 4S battery to maintain electrical isolation and safety compliance. Flight control is managed by a Pixhawk 6C running PX4 firmware, while autonomous computation is handled by an onboard Raspberry Pi 5 that interfaces with the flight controller using the MAVLink protocol. Mission planning and monitoring are conducted through QGroundControl, with autonomous functionality implemented in Python to include waypoint navigation, image processing, object detection, LiDAR processing, and payload delivery.

The airframe integrates 6061-T6 aluminum structural members and motor mounts with carbon fiber arms to balance stiffness and mass. Payload manipulation is achieved through an electromagnet system controlled by a custom circuit receiving GPIO commands from the Raspberry Pi. When energized, the magnet enables passive pickup by hovering over the payload, eliminating mechanical latching mechanisms. This design reduces moving parts, lowers mechanical failure risk, and enables fast, repeatable resupply while improving deployment reliability and reducing the likelihood of payload misalignment or shatter events.

2 Team & Schedule Overview

Our team is organized to reflect a professional flight program structure with clear program, technical, and flight authority. The Project Manager oversees schedule and deliverables, the Chief Engineer is responsible for overall airworthiness, and the Pilot in Command leads flight operations. The Faculty Advisor provides oversight on safety and formal design reviews.

2.1 Team Organization

Member	Role	Email	Class
Ethan Stewart	Project Manager	e.t.stewart@wustl.edu	Junior
Diego Perez	Chief Engineer	d.s.perez@wustl.edu	Sophomore
Teo Giannini	Chief Electrical Engineer & Pilot in Command	a.giannini@wustl.edu	Sophomore
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Elias Terman	Backup Pilot & Test Lead	e.p.terman@wustl.edu	Freshman
Jeff Krampf	Faculty Advisor	jkrampf@wustl.edu	Faculty

2.2 Schedule Overview

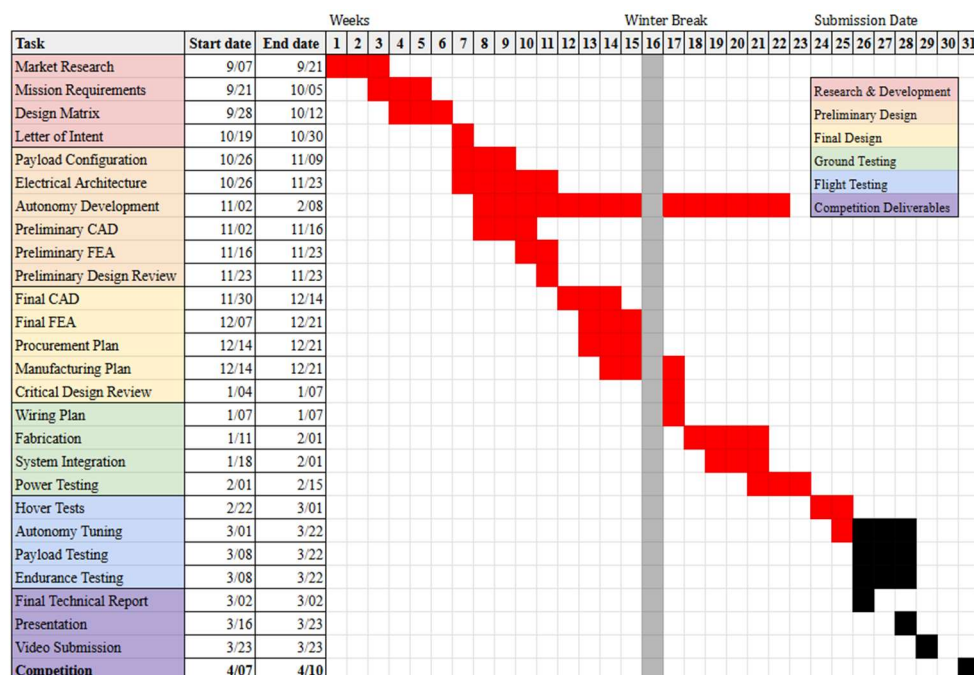


Figure 1: Gantt Chart

3 Q1: What is your aircraft, and why did you design your aircraft this way?

3.1 Identifying Mission Requirements

Requirement	Category	Descriptions
Drone complies with General FAA Laws & Part 107 practices along with DBVF Flight Safety standards	Safety	Keeps spectators and all participants of VFS DBVF Competition safe at all times
Overall weight must be below 25 lbs	Mission Parameter	Maximize points during FM2 & FM3, conform to competition guidelines
Aircraft may not exceed 8" x 8" x 8".	Mission Parameter	Maximize points during FM2 & FM3, conform to competition guidelines
Payloads must survive a drop of 30 feet	Mission Parameter + Safety	Maximize points during FM2 & FM3, conform to competition guidelines
Payloads must weigh at least 0.5 lbs. A maximum of 5 payloads may be attached to one bundle.	Mission Parameter	DBVF allows multiple payloads to be picked up at once, we will target a multi-bag payload to balance efficiency, loading time, and consistency during flight
Fly for 10 minutes with max payload.	Mission Parameter + Aircraft Design (Range)	FM-2 & FM-3 are timed competitions in which our aircraft must be able to drop off as many payloads into Fire 1 or Fire 2. Having a 10 minute flight time will provide enough time to complete each mission.
Maintain hover in 9 mph winds (average crosswind in Churchville, MD).	Safety + Mission Parameter	Aircraft will be subject to windspeeds averaging 9mph in Churchville, so the VTOL must be able to hover and complete FM's 1-3 in these conditions
Hover in place for 5+ seconds.	Mission Parameter + Aircraft Design (Range)	Maximize points during GM1, conform to competition guidelines
Total cost must be under \$8650	Build Cost/Budget	Funding was secured for our design team through university and student union project grants, projected total cost of aircraft
The aircraft must achieve a high payload pickup/drop-off accuracy.	Aircraft Design - Payload	Payload drop accuracy was defined as a critical quantitative metric because of the tight target zone and its role as a scoring multiplier.
Propellers blades must be a minimum of 3 inches apart from tip to tip	Aircraft Design - Performance	To minimize air disturbance between propeller blades, there must be a minimum of 3 inches between blades. This value was found from various manufacturers.

Figure 2: Mission Requirements

Our targeted requirements were defined as quantifiable design goals and organized into four categories: safety, mission parameters, aircraft design, and build cost/budget. Overall, the requirements were developed from DBVF competition rules and supplemented by conventional VTOL design practices informed by industry consultation and team experience.

3.2 Optimized Configuration Design Matrix

Criteria:	Size	Payload Capacity	Build Cost	Performance	Stability	Range	Total Scores
Weight:	1	2	2	1.5	2	1.5	Max = 100
Quadcopter	6	5	7	6	6	5	58.5
Coaxial Quadcopter	6	6	5	7	7	4	58.5
Hexacopter	5	7	6	7	8	5	65
Octocopter	4	8	5	6	9	4	63
4-Engine Quadplane	4	6	4	8	6	8	60
8-Engine Quadplane	3	7	3	8	7	7	59.5

Figure 3: Decision Matrix

A weighted decision matrix was used to evaluate potential aircraft configurations based on key performance metrics. Designs were scored on a 1–10 scale, with 1 representing unsatisfactory performance and 10 representing exceptional performance. Designs requiring lift surfaces, phase transitions, or rotating motors were deprioritized due to increased complexity and limited benefit. Because the mission emphasizes vertical lift and multi-payload transport over short distances, configurations prioritizing vertical thrust and hover stability were weighted most heavily.

3.3 Propulsion

Propulsion system sizing was driven by thrust margin, hover efficiency, and compatibility with 6S electrical architecture. The aircraft uses six T-Motor MN6007 II (320KV) paired with 21×6.3-inch carbon fiber propellers, powered by a 6S 12,000mAh lithium polymer battery. The 320KV rating was selected to match large diameter, low RPM propellers on 6S voltage, prioritizing hover efficiency and maximum flight time.

At the 25lb (11,340g) maximum takeoff weight, required hover thrust per motor:

$$T_{motor} = \frac{mg}{6} = \frac{11340}{6} \approx 1890g$$

Manufacturer data shows that at 47%, throttle each motor produces 1926g of thrust at 3380RPM while drawing 196W, corresponding to an efficiency of 9.84g/W. Since $1926g \approx 1890g$, hover occurs at approximately 47% throttle, placing operation within the motor's peak efficiency band.

Total hover power:

$$P_{ho} = 6 \times 196 \text{ W} = 1176 \text{ W}$$

Battery energy is calculated from nominal voltage and capacity:

$$E = V \times Ah = 22.2 \text{ V} \times 12 \text{ Ah} \approx 266 \text{ Wh}$$

Estimated hover endurance is therefore bound by:

$$t = \frac{E}{P_{hov}} = \frac{266}{1176} \approx 0.226 \text{ hr} \approx 13.5 \text{ minutes}$$

Accounting for 80% usable battery capacity to preserve LiPo longevity, practical hover endurance is approximately 10.8 minutes, still exceeding the 10-minute mission requirement.

Maximum thrust per motor is 5397g at 100% throttle. Total available thrust:

$$T_{total,max} = 6 \times 5397 = 32,382 \text{ g}$$

System thrust-to-weight ratio:

$$\frac{T}{W} = \frac{32382}{11340} \approx 2.86$$

This provides substantial maneuverability and wind correction ability while ensuring hover occurs well below peak loading, reducing motor stress and improving efficiency.

3.4 Payload Mechanism

During preliminary payload development, four primary requirements were established: structural integrity under load, non-interference with onboard systems, multi-payload release capability,

and preservation of flight stability. Based on testing and subsystem evaluation, electromagnetic pickup and release architecture was selected for its simplicity, reliability, and manufacturability. Testing showed that minimizing moving components significantly improved system robustness and reduced integration risk with the aircraft avionics and structure.

The final mechanism consists of a cylindrical arm extending from the fuselage with an electromagnet mounted at its base, interfacing with a circular steel plate integrated into a detachable “starfish” payload basket capable of carrying up to five sandbags. By eliminating mechanical latches and minimizing moving components, the design reduces potential failure modes while maintaining consistent payload alignment. The maximum operational payload is 2.5lb, while the electromagnet is rated at 10lb, yielding a factor of safety of 4. Even under a 3g vertical acceleration event corresponding to maximum thrust, the effective holding force remains above the required retention load.

Structural components were 3D printed in PETG after material testing confirmed superior impact resistance and rigidity compared to PLA and ABS. Optimized print parameters reduced total mechanism mass to 0.88lb while maintaining required stiffness and strength. Each payload bundle also incorporates an ArUco marker for visual identification, allowing a Raspberry Pi camera and onboard computer vision system to locate and automatically pick up payload bundles, without additional mechanical complexity.

3.5 Electronics

The electrical system is divided into two fully independent power domains: propulsion and avionics, ensuring high current motor loads are isolated from flight critical electronics to maintain stable voltage and fault prevention. Propulsion is powered by a 6S 12,000 mAh LiPo battery distributed to six individual 70A continuous ESCs, each driving one motor. Individual ESCs were selected over a multi-in-one unit to improve thermal distribution, simplify maintenance, and eliminate single point motor failure risks. Propulsion power can be physically disconnected via XT90 battery removal and externally controlled through a kill plug and remote disarm capability.

Flight control is managed by a Pixhawk 6C running PX4 firmware, selected for its compact form factor, extensive I/O, and strong compatibility with Python and MATLAB software stacks. The flight controller performs real time stabilization and motor control, while a Raspberry Pi 5 companion computer executes higher level autonomy logic, including ArUco-based payload identification and mission management beyond core stabilization. A communications cutoff switch integrated into the avionics power system can disable the VTX, receiver, and telemetry radio without affecting propulsion or flight critical systems. Combined with firmware-configured failsafes and geofencing constraints, this layered safety design ensures controlled responses to signal loss, manual override, or subsystem faults while preventing failure propagation between propulsion, payload, and flight control electronics.

4 Drawing Package

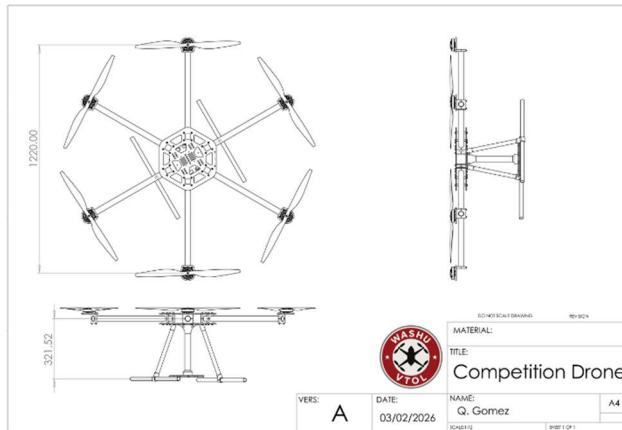


Figure 4: Competition Drone Standard 3-View

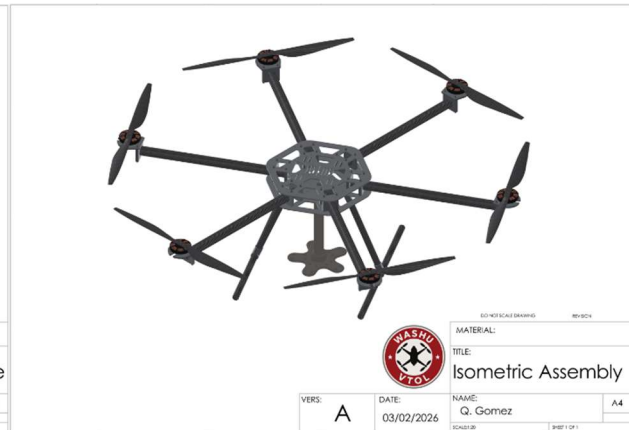


Figure 5: Competition Drone Isometric View

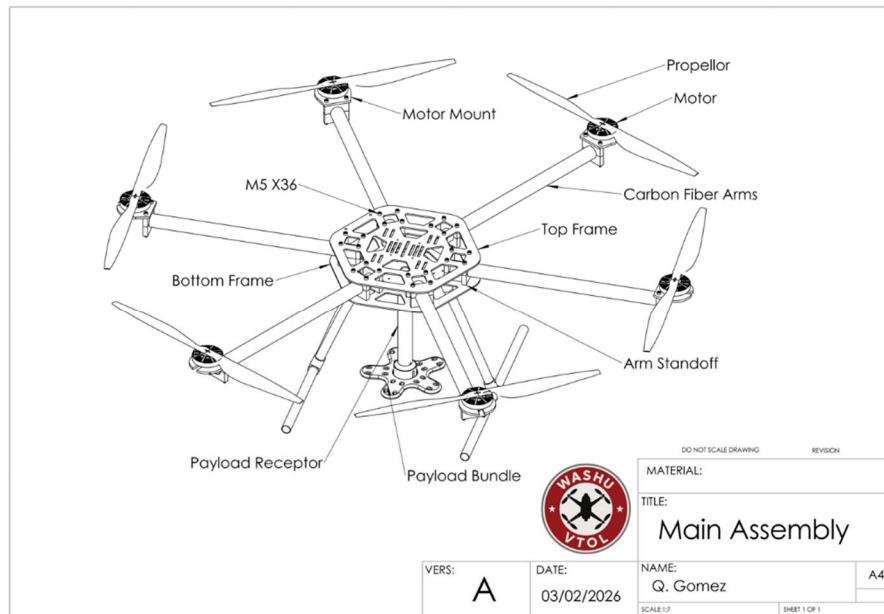


Figure 6: Competition Drone Labeled Isometric View

Components	Measurements (mm)
Propeller Diameter	530
Top/Bottom Frame Length	335
Top/Bottom Frame Thickness	3.175
Total Frame Height	56.35
Payload Receptor Length	280
Carbon Fiber Arms Length	495
Carbon Fiber Arms Diameter	30
Total Wingspan	1220
Motor Mount Thickness	5
Arm Standoff Height	50
Arm Standoff Width	25
Arm Standoff Thickness	10

Figure 7: Key Component Measurements

5 Q2: How did you build, test, and validate your aircraft?

5.1 Fabrication Methods

For the top and bottom frame plates, 6061-T6 aluminum, 7075-T6 aluminum, and carbon fiber composite plates were evaluated. Although 7075-T6 provides higher strength, required safety factors were achieved with 6061-T6 at a thickness of 3.175 mm (1/8 in), making the added cost of 7075 unnecessary. Carbon fiber plates were not selected due to higher cost and more complex manufacturing. The plates were cut using waterjet for primary geometry and CNC drilled for precise hole alignment and assembly accuracy.

The arm clamps were designed as two split halves that bolt together to compressively engage with a 30mm carbon fiber arm, providing a pressure fit. Arm clamps and motor mount top plates were also evaluated in 6061-T6 aluminum, 7075-T6 aluminum, and 3D-printed ULTEM 1010. 6061-T6 aluminum was selected because it provides sufficient strength and stiffness without the cost premium of 7075. Polymer options were not used for these parts due to concerns with long-term stiffness and load transfer. These components were CNC machined and outsourced to meet schedule constraints and ensure dimensional accuracy. The arms were fabricated from carbon fiber tubing instead of aluminum tubing due to the improved strength-to-weight ratio. Reducing arm mass lowers bending moments at the frame interface and decreases overall structural loading. The carbon fiber 3K weave provides fiber alignment along the primary load paths and conforms well to tubular geometries, reducing the likelihood of stress concentrations.

Non-structural components, including avionics trays, payload mounts, and wire management brackets, were 3D printed to allow rapid iteration and easy replacement. For the payload mechanism, PLA, ABS, and PETG were considered. PETG was selected for its balance of rigidity and impact resistance, which is beneficial during repeated drop operations. Structural fasteners consist of M5 bolts with locknuts to maintain preload and improve reliability under vibration. Fastener sizing was selected to ensure adequate shear and tensile capacity under maximum thrust conditions.

5.2 Validation of Design Choices

Physical inspection following assembly confirmed proper fitment, alignment, and mounting rigidity of the frame. The propulsion system sizing was validated through static thrust testing and motor direction confirmation. Hover endurance and power predictions will be validated using onboard telemetry logs collected during full weight flight testing. Payload bundle reliability, propulsion redundancy behavior, autonomous mission stability, and failsafe performance will be confirmed through repeated mission profile testing. Autonomy readiness and payload reliability will be demonstrated through consecutive full cycle mission runs to verify repeatability under operational conditions. These validation efforts are intended to confirm the design trade decisions favoring hexacopter architecture using large diameter, low RPM propellers, and electrically isolated propulsion and avionics subsystems for wildfire mission optimization.

Finite element analysis was conducted on the motor mounts and carbon fiber arms under maximum thrust loading conditions. Results indicate a factor of safety of approximately 11 for both the arm structure and the motor mount assemblies under peak thrust loads. These margins confirm that the primary load bearing components remain well below material yield limits during worst case thrust events, providing structural robustness while maintaining acceptable mass efficiency.

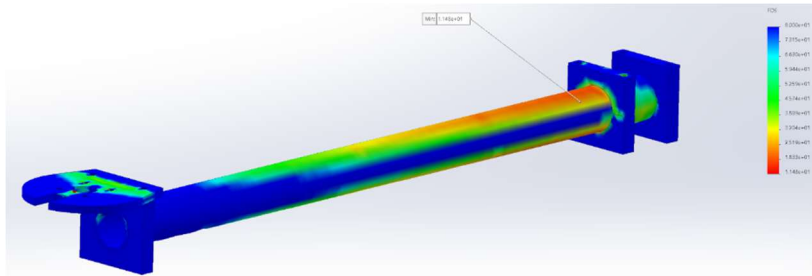


Figure 8: Arm FOS

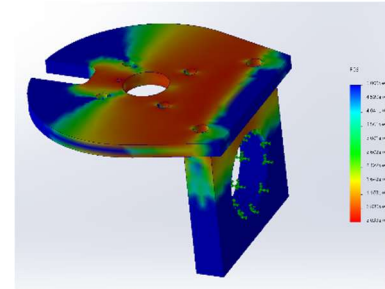


Figure 9: Motor Mount FOS

5.3 Ground Testing

Prior to any flight operations, a staged ground testing program was conducted to validate structural integrity, electrical safety, voltage stability, and subsystem reliability. Initial electrical integration was performed using a resettable “smoke stopper” fuse device to limit current during first power on, preventing component damage from potential wiring faults. As components were added, corresponding voltage rails were verified before connection. Following full electrical integration and frame assembly, detailed inspections confirmed secure mounting hardware, proper wire routing, and absence of structural deformities.

Safety critical systems were then configured and validated, including the external kill switch, LiPo battery connections, auxiliary pre-arm and propulsion kill switches, and firmware-based failsafe protocols. Motor parameters in the AM32 configurator were cross-checked against physical motor directions to ensure correct rotation. The aircraft was rigidly mounted to a fixed test stand, and throttle response was verified first without propellers and then with propellers installed for thrust characterization at hover representative throttle levels. Static thrust results met the required lift capability to support the 25lb maximum takeoff weight while ESC temperature rise was monitored during sustained ground runs to confirm acceptable thermal margins.

The electromagnetic payload mechanism was tested using a mock payload to verify reliability, release consistency, and resistance to vibration. Electrical isolation between propulsion and avionics power domains was validated to ensure avionics and payload systems remain powered during propulsion disarm events. Finally, comprehensive sensor calibration and signal validation were performed, including GPS hold verification, IMU calibration, LiDAR functionality checks, and Raspberry Pi camera visibility confirmation. Only after successful completion of these incremental validation steps was the aircraft approved for flight testing.

5.4 Flight Testing

Due to unforeseen university funding delays, delivery of propulsion components and hardware procurement was postponed, compressing the available flight-testing window. To prepare for flight testing, a structured and incremental validation plan has been implemented. Initial free hover and low speed translation tests to validate stability, yaw authority, and performance across increasing payload weights. Manually controlled drop tests to verify release timing and accuracy, followed by manual and payload pickup validation to confirm engagement reliability.

Autonomy testing will be conducted in stages before full integration. The drone will first be manually hovered over a test tarp containing printed ArUco markers to evaluate the image-processing pipeline, including the edge detection process and ArUco detection. After testing the vision system in isolation, integrated flight tests will be conducted. The aircraft will be commanded to descend over both marked and unmarked tarps to evaluate system performance, after which parameters will be tuned to improve final performance.

Final testing will include full autonomous execution of FM-2 and FM-3 mission profiles using PX4 mission planning to demonstrate continuous autonomous payload cycling. Mission simulations will also be conducted across varied weather inputs and system failure scenarios to validate robustness prior to physical testing.

5.5 Software Testing

As pre-flight software testing, the use of the Pixhawk 6C PX4 based flight controller and ArUco codes was simulated with Gazebo. A built-in template of a quadcopter with a downfacing camera was used within gazebo for the simulation use case. Despite Gazebo only having a quadcopter template, functionality of this simulation is not affected. Python code with the MavSDK library is used for connection to the flight controller, and the OpenCV library for ArUco recognition.

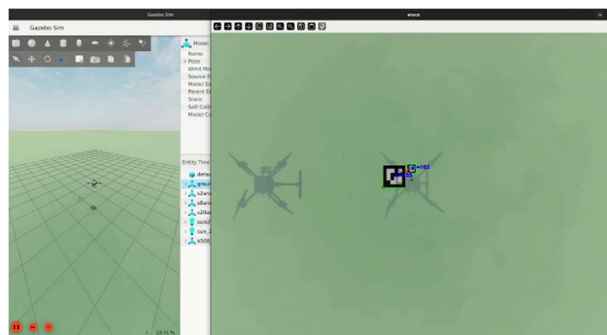


Figure 10: Gazebo simulation of quadcopter with ArUco code recognition and distance approximation (155cm and 163 cm).

The tarp edge detection and recognition program was tested, but due to not having flown yet, test reference images could not be taken. Therefore, generative AI was used to create simulated images and videos of tarps in various environments and lighting conditions. This would allow for testing before flight, and with a wide variety of conditions.

6 Q3: What is innovative about your aircraft, and why does this matter for the competition's focus on wildfire response?

6.1 Innovation 1 – Payload System

For bushfire suppression missions, reliability, stability, and precision delivery are essential when targeting small ignition points. The aircraft's payload delivery system is innovative because it replaces conventional mechanical drop mechanisms with a fully integrated electromagnetic system (EMS). By eliminating servos, linkages, and rotating assemblies, the EMS removes common failure modes associated with moving components, particularly in high temperature and turbulent wildfire environments. Payload release is achieved by de-energizing the magnet, allowing for rapid deployment without mechanical delays. Additionally, the single point magnetic attachment interface ensures repeatable, centered engagement with the aircraft. This prevents off axis loading and guarantees consistent alignment during autonomous pickup and release operations. The result is a payload system that is mechanically simple, structurally robust, aerodynamically balanced, and tightly integrated with onboard computation.

The EMS is directly integrated with the Raspberry Pi and flight control system, enabling release to be triggered algorithmically using GPS position, ground speed, and wind correction calculations. This allows suppressant deployment to be synchronized precisely with aircraft state rather than relying solely on manual timing. For wildfire response, where crosswinds and small target areas can significantly affect drop accuracy, this software driven release capability improves repeatability and precision.

The payload bundle itself is implemented as a rigid, circular five-point bundle designed to maintain symmetric mass distribution during all phases of flight. This configuration ensures a consistent center of gravity and eliminates incremental mass shifts that could destabilize the aircraft. Circular geometry distributes loading evenly around the magnetic interface, minimizing torsional imbalance and preserving predictable aerodynamic behavior. Structural validation was conducted through finite element analysis and impact testing, including 40 ft drop tests on concrete to confirm durability beyond expected operational conditions. This testing ensures the bundle maintains structural integrity during deployment and ground impact in rugged terrain.

This innovation adds value to wildfire missions by increasing deployment accuracy, reducing mechanical failure risk, and maintaining stable flight characteristics during suppressant delivery. By combining structural symmetry with software triggered electromagnetic release, the aircraft delivers a precision focused solution specifically tailored to early-stage fire containment and rapid response bushfire suppression.

6.2 Innovation 2 – Autonomous Software Stack

To implement autonomous control for the missions and payload pickup, a multi-stage software pipeline was developed to separate autonomous descent into distinct operational phases. The

system first navigates to the approximate pickup or drop location using GPS waypoint guidance. Once within the target area, the image processing transitions to a horizontal alignment phase, where a tuned combination of Gaussian blur and edge detection is used to identify tarp boundaries and compute a centroid for centering (done with OpenCV library). This approach allows the aircraft to more precisely center itself in the target area even without fiducial markers.

When ArUco codes are present before descent (during payload pickup), ArUco locating is prioritized over edge detection to provide more accurate centering relative to the payload. This prioritization logic ensures that higher accuracy visual data is used when available, while keeping a fallback strategy when there aren't markers. In practice, our design implements three different ArUco codes during flight missions: 20cm x 20cm, 8cm x 8cm, and 2cm x 2cm. The biggest code is used to find the payload placement location with a GPS margin of error of 2m, the middle code is for smooth transitioning to the smallest, and the smallest code is used to get as close as needed (down to 3cm). With the ArUco image processing, each code provides its distance to the camera, and angle of the code on a plane. This is then combined with the LiDAR sensor readings to get the exact relative position of the drone from the package

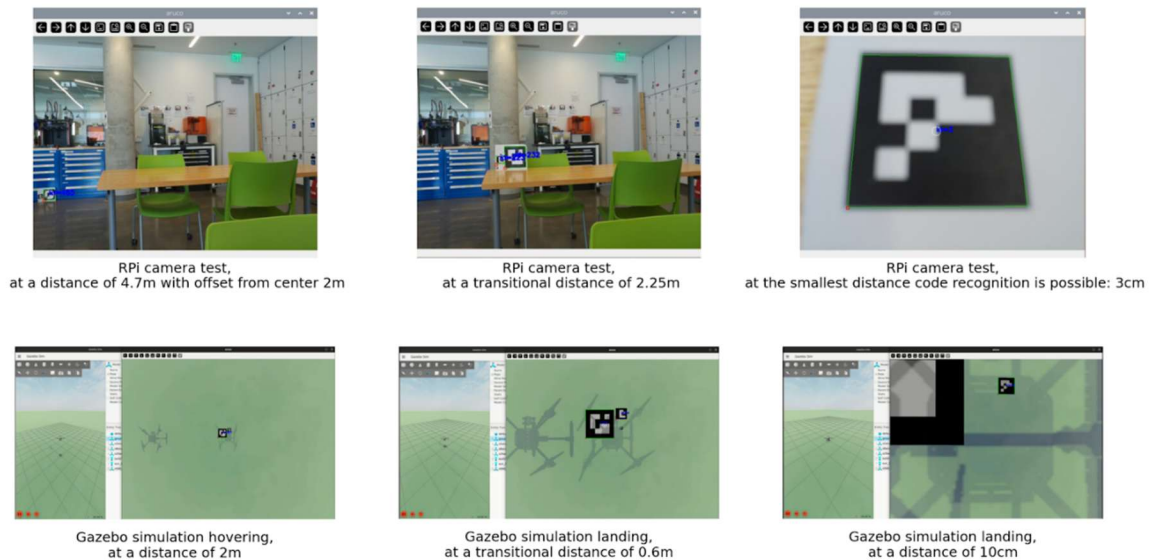


Figure 11: Comparison between real-world Raspberry Camera (RPI) images and simulated images at different distances.

During descent below 30 ft AGL, altitude measurements switch from using the barometer provided by the flight controller, to LiDAR measurements. This transition improves vertical accuracy during the most critical phase of the operation, where small altitude errors can significantly affect pickup consistency. The sensor switching threshold was chosen to balance reliability from the barometer at higher altitudes with LiDAR precision near the ground.

By structuring navigation sequentially with GPS guidance, image processing phases, and LiDAR assisted descent phases, the system has redundancy and doesn't solely rely on one sensing

mechanism. This also improves performance under varying lighting conditions, GPS uncertainty, and environmental disturbances, resulting in more consistent payload pickup and drop performance. For the wildfire response mission, this autonomy architecture enables faster and more reliable payload deployment and resupply operations, reducing the need for manual piloting and allowing the aircraft to repeatedly deliver fire-suppressant payloads with high positional accuracy. The improved reliability and precision of autonomous pickup and drop cycles increases mission effectiveness and allows more suppressant deliveries within the limited mission time window, directly impacting wildfire containment efforts.

6.3 Innovation 3 – Custom Electromagnet Control Circuit

The payload pickup mechanism is a simple winding electromagnet matched to the required holding force and power constraints. However, directly powering the magnet from a shared avionics rail brought up risk to flight critical electronics due to voltage spikes and transient current draw. To address this, a dedicated electromagnet control circuit was designed and integrated into the avionics system.

The circuit is controlled from a GPIO on the onboard computer and is powered by an independent BEC to isolate the system from the main flight controller power rails. This isolation prevents voltage sag or inductive kickback from propagating into important sensors and flight control during activation or release. Protection and gating logic were implemented using an N-type MOSFET for switching, a flyback diode for transient suppression, and passive filtering components including capacitors and resistors to stabilize both the supply rail and control signal. All components were assembled onto a perf board and secured within the airframe with strain relief to prevent mechanical stress on solder joints.

The circuit was tested under full load holding conditions while monitoring the regulated 5V power to confirm that no voltage drop or reset behavior happened during activation. This testing confirmed effective isolation and stability. By separating electromagnetic power from other avionics and implementing hardware protection, the system converts a simple electromagnet into a controlled, electrically isolated system. This architecture directly benefits the wildfire response mission by ensuring that payload pickup and release events cannot interfere with navigation sensors, flight control, or autonomous guidance systems during critical mission phases.

6.4 Why This Matters for Wildfire Response

In real wildfire response, effectiveness is measured by how quickly and precisely suppressant can be delivered and redelivered to evolving fire fronts. WashU VTOL's innovations directly target this metric by enabling rapid autonomous payload cycling, high precision low altitude drop accuracy, reduced operator workload and human-error risk, mission continuity during electrical or propulsion faults. Together, these innovations transform the aircraft from a single drop reconnaissance drone into a repeatable, autonomous, high throughput wildfire suppression platform, closely mirroring real first-response wildfire containment efforts.

7 Reflection & Next Steps

7.1 Reflection

The primary technical challenge encountered during development was the payload system, particularly its integration with avionics and compliance with competition safety rules. While the electromagnetic concept was mechanically simple, ensuring proper electrical isolation and reliable execution through the flight controller required extensive coordination between mechanical and electrical subteams. The interface between the magnet driver, Raspberry Pi, and flight controller introduced constraints that influenced wiring layout, power budgeting, and failsafe logic. This reinforced the importance of early cross disciplinary integration rather than sequential subsystem development. Designing within these constraints required iterative redesign and closer review of the rulebook than initially anticipated.

Manufacturing time was also underestimated. Additive manufacturing of structural and payload components required multiple print iterations, post-processing, and fit adjustments. Lead times for printing, reprinting after minor geometry changes, and coordinating part availability slowed integration more than expected. This emphasized the importance of early prototype fabrication, realistic manufacturing scheduling, and building margin into the development timeline.

Overall, the key lesson from this first year has been the necessity of systems level thinking. Effective design required coordination across structures, propulsion, avionics, and competition constraints, supported by disciplined iteration and documentation.

7.2 Next Steps

The next phase focuses on structured flight testing and incremental validation. Ground checks will verify propulsion response, structural integrity, authority control, and sensor calibration prior to flight. Testing will then progress from low altitude hover to maneuvering, payload validation, and full mission execution using a defined test matrix. Pilot training will occur in parallel to ensure consistent handling and emergency response proficiency. All flights will be logged and reviewed, with deviations addressed through control tuning, vibration mitigation, or minor mechanical refinements. Configuration control will be maintained for all updates.

Remaining work includes final autonomy tuning for wind corrected drop timing, endurance validation under realistic battery discharge conditions, and repeated mission runs to demonstrate reliability. Crosswind hover testing and aggressive ascent/descent profiles will validate environmental robustness. Competition readiness will be declared only after consistent, fault free mission completion, repeatable payload accuracy, and full verification of geofencing, fail safes, and remote disarm functionality.

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